

End-to-End Test Scenarios

Scenario catalogue for Purchase Order Preparation

Purpose. Test the complete user journey from input data to decision, exception handling, PO draft, approval state and audit evidence.

Test coverage map

	Input	Decision	Qty	Commercial	Exception	PO	Approval	Audit
Happy path	✓	✓	✓	✓	✓	✓	✓	✓
No reorder	✓	✓						✓
MOQ / pack	✓	✓	✓			✓		✓
Open PO offset	✓	✓	✓			✓		✓
Missing supplier	✓			✓	✓	✓		✓
Missing price	✓			✓	✓	✓		✓
Supplier not approved	✓			✓	✓	✓		✓
Multiple suppliers	✓			✓		✓		✓
Multiple lines same supplier	✓			✓		✓		✓
Critical edit after approval						✓	✓	✓
Duplicate draft request						✓		✓
Mixed valid + blocked	✓	✓	✓	✓	✓	✓		✓
Rerun / idempotency	✓					✓		✓
Traceability / explanation		✓		✓	✓	✓	✓	✓
Performance / usability	✓	✓				✓	✓	✓

1. Test design approach

The test set is deliberately broader than module-level business-rule tests. Each scenario starts from an input state and follows the workflow to a visible user outcome: decision, exception, PO draft, approval state and evidence trail.

2. Core scenario catalogue

ID	Scenario	Input setup	Expected outcome	Type
E2E-01	Happy path - one reorder SKU	Valid stock / demand / open PO; supplier and price available	Correct line prepared; waits for human approval	Normal
E2E-02	No reorder required	Available position already covers	No PO line; reason shown	Normal

ID	Scenario	Input setup	Expected outcome	Type
		requirement		
E2E-03	Several valid SKUs, same supplier	Multiple reorder SKUs map to same supplier	One draft contains all valid lines for that supplier	Workflow
E2E-04	Several suppliers	Valid reorder SKUs map to different suppliers	Separate drafts / groups by supplier as designed	Workflow
E2E-05	MOQ adjustment	Raw requirement below defined MOQ	Final proposed quantity respects expected MOQ answer	Calculation
E2E-06	Pack multiple adjustment	Raw requirement not divisible by pack multiple	Final proposed quantity matches expected rounded answer	Calculation
E2E-07	Open PO reduces need	Open PO exists for the SKU	Decision reflects the agreed quantity logic	Calculation
E2E-08	Missing supplier mapping	Reorder required; no default / qualified supplier	SKU blocked and kept visible	Exception
E2E-09	Supplier not approved	Supplier record exists but approved flag is No	Approval blocked; clear reason	Exception
E2E-10	Missing price	Supplier exists; unit price blank / invalid	Blocking exception; no fabricated price	Exception
E2E-11	Mixed valid + blocked SKUs	At least one valid and one blocked SKU	Both outcomes remain visible; partial-progress behaviour follows team decision	Workflow
E2E-12	All SKUs blocked	Every candidate has a blocking commercial issue	No false “success”; exception summary visible	Exception
E2E-13	Line amount mismatch	Displayed amount differs from qty × price	Mismatch corrected / approval blocked	Control
E2E-14	PO total mismatch	PO total differs from sum of line amounts	Approval blocked until consistent	Control
E2E-15	Critical edit after approval	Change supplier, quantity or unit price	Status returns to Needs Review	Control
E2E-16	Non-critical review note	Add reviewer comment only	Approval state follows intended design	Control
E2E-17	Duplicate draft	Repeat same run +	No second active draft	Control

ID	Scenario	Input setup	Expected outcome	Type
	request	supplier generation		
E2E-18	Input order changed	Same records in different row order	Same business outcome	Robustness
E2E-19	Repeat run with unchanged inputs	Run the exact same case again	Deterministic result; no hidden duplication	Robustness
E2E-20	Correct missing price and retry	Resolve a blocked price then rerun / continue	Exception clears and valid line proceeds	Recovery
E2E-21	Correct supplier approval and retry	Resolve supplier status	State updates consistently	Recovery
E2E-22	Reviewer edits quantity	Human changes proposed quantity before final approval	Dependent checks rerun; change is visible	Human-in-loop
E2E-23	Reviewer holds draft	User does not approve	Draft remains unapproved and clearly labelled	Human-in-loop
E2E-24	Traceability check	Select any prepared PO line	Source / replenishment reference can be identified	Audit
E2E-25	Reviewer identity / timestamp	Approve a clean draft	Identity and timestamp captured	Audit
E2E-26	Ten-SKU batch	Full prototype dataset	All SKUs reach a visible state; run finishes reliably	Performance
E2E-27	High exception mix	Several different blocker types in one run	Exceptions are distinguishable and actionable	Usability
E2E-28	Explanation consistency	Compare reason text for similar decisions	Same rule produces consistent explanation	Usability
E2E-29	Manual-vs-agent paired case	Same scenario run both ways	Agent result matches expected answer with lower effort	Evaluation
E2E-30	Interrupted review and resume	Stop after draft generation, then return	State is recoverable without silent loss / duplicate	Recovery

3. Metamorphic tests

These checks test relationships between runs rather than a single fixed answer. They are useful for catching behaviour that a normal happy-path test may miss.

Metamorphic check	Expected relation
Row-order invariance	Reorder the same input rows. Business result should not change.
Idempotent rerun	Repeat the same generation request. A second active draft should not appear.
Irrelevant-field stability	Change a descriptive field that should not affect quantity. Decision should stay the same.
Critical-field sensitivity	Change supplier, quantity or unit price after approval. Review state should change.
Exception resolution	Fix exactly one missing field. Only the related blocker should clear.